MAST30027 Modern Applied Statistics

Exam Problem Sheet

Fresh exam-style problems for every week of MAST30027, built from the lectures, tutorials, extra notes and the question patterns of the 2016–2022 papers (none of which are reproduced). Everything is by hand: R output is printed for you to read, never to write. Marks follow the 2-hour, 50-mark format. Full worked solutions are in Part II.

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Part I Problems

Week 1: MLE, Fisher information, asymptotics

This is Q1 on every recent paper (7–10 marks). The course's $I(\theta)$ is the information of the *whole sample*, $I(\theta) = -\mathrm{E}[\ell''(\theta)]$, and the answer to "asymptotic distribution" is $\hat\theta \approx \mathrm{N}\big(\theta, I(\theta)^{-1}\big)$.

Problem 1.1 (8 marks)

Let $X_1, \dots, X_n$ be iid Rayleigh with density

$$f(x;\theta) = \frac{x}{\theta}\exp\!\left(-\frac{x^2}{2\theta}\right), \qquad x>0,\ \theta>0,$$

and you may use $\mathrm{E}(X^2) = 2\theta$.

(a) Write down the log-likelihood $\ell(\theta)$.

(b) Find the Fisher information $I(\theta)$.

(c) Find the MLE $\hat\theta$ and state its asymptotic distribution.

(d) Show that the observed information $-\ell''(\hat\theta)$ equals $I(\hat\theta)$ for this model, and explain in one or two sentences why we are allowed to plug $\hat\theta$ into $I(\theta)$ to get a standard error.

Problem 1.2 (7 marks)

Let $X_1, \dots, X_n$ be iid with density $f(x;\theta) = \theta x^{\theta-1}$ for $0<x<1$, $\theta>0$.

(a) Find $\ell(\theta)$, the MLE $\hat\theta$ and the Fisher information.

(b) A sample of $n=20$ gives $\sum \log x_i = -12.5$. Compute $\hat\theta$, its standard error and an approximate 95% Wald confidence interval for θ.

(c) Using the invariance property, give the MLE of $\mathrm{E}(X) = \theta/(\theta+1)$.

Problem 1.3 (8 marks, the "data in ℓ''" twist)

Let $X_1, \dots, X_n$ be iid Gamma with known shape 3 and unknown mean m, so that

$$f(x;m) = \frac{27}{2m^3}\,x^2\,e^{-3x/m}, \qquad x>0, \qquad \mathrm{E}(X)=m.$$

(a) Write down $\ell(m)$ and find the MLE $\hat m$.

(b) Show that $\ell''(m)$ depends on the data, then find $I(m)$.

(c) State the asymptotic distribution of $\hat m$ and compare it with the exact variance of $\bar X$.

Problem 1.4 (7 marks)

Let $X_1, \dots, X_n$ be iid Inverse-Gaussian with known mean 1 and unknown shape $\lambda>0$:

$$f(x;\lambda) = \sqrt{\frac{\lambda}{2\pi x^3}}\exp\!\left(-\frac{\lambda(x-1)^2}{2x}\right), \qquad x>0.$$

(a) Find $\ell(\lambda)$, the MLE $\hat\lambda$ and $I(\lambda)$.

(b) State the asymptotic distribution of $\hat\lambda$.

(c) With $n=50$ and $\sum (x_i-1)^2/x_i = 12.5$, give $\hat\lambda$ and an approximate 95% CI.

Week 2: Binomial regression

Model: $Y_i \sim \text{Bin}(m_i, p_i)$, $g(p_i) = \eta_i = x_i^T \beta$ with logit, probit or cloglog link. Exam skills: likelihood and score by hand, odds-ratio interpretation, and a prediction CI built on the η scale then transformed.

Problem 2.1 (8 marks)

Let $Y_i \sim \text{Bin}(m_i, p_i)$ independently for $i = 1, \dots, n$, with $\text{logit}(p_i) = x_i^T \beta$.

(a) Write the log-likelihood as a function of β (drop constants).

(b) Show that the score is $\partial \ell / \partial \beta_j = \sum_i x_{ij}(y_i - m_i p_i)$.

(c) Show that the Fisher information matrix is $X^T W X$ with $W = \text{diag}\big(m_i p_i (1 - p_i)\big)$, and explain why the observed and expected information coincide for the logit link.

(d) Write down the asymptotic distribution of $\hat{\beta}$ and say how you would get a standard error for $\hat{\beta}_j$.

Problem 2.2 (10 marks)

A seed-coating trial sowed 16 batches of 50 seeds. Each batch was kept at a temperature `temp` (12–26 °C) and was either coated or not (`coat`, levels `no`, `yes`). The number germinated is `germ`.

```
> mod1 <- glm(cbind(germ, 50 - germ) ~ temp + coat, family = binomial, data = seeds)
> summary(mod1)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept) -5.87928    0.44669 -13.162  < 2e-16 ***
temp         0.30714    0.02254  13.627  < 2e-16 ***
coatyes      0.85723    0.17635   4.861 1.17e-06 ***
---
(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 294.33  on 15  degrees of freedom
Residual deviance: 11.80  on 13  degrees of freedom
AIC: 79.18

> predict(mod1, newdata = data.frame(temp = 18, coat = "yes"),
+         type = "link", se.fit = TRUE)
$fit
        1
0.5064256
$se.fit
[1] 0.1241462
```

(a) Interpret the coefficient of `coatyes` as an odds ratio and give a 95% CI for that odds ratio.

(b) By what factor do the odds of germination change when temperature rises by 5 °C, coat held fixed?

(c) Using the `predict` output, estimate the germination probability for a coated seed at 18 °C and give a 95% CI for it.

(d) Explain why the CI in part (c) is built on the link scale and then transformed, rather than as $\hat{p} \pm 1.96 \times \text{SE}(\hat{p})$.

(e) At what temperature does the model predict a 50% germination rate for coated seeds?

Problem 2.3 (4 marks)

For one batch the fitted linear predictor is $\hat{\eta} = 0.4$.

(a) Give $\hat{p}$ under the logit, probit and complementary log-log links.

(b) Which of these links is not symmetric about $p = 0.5$, and what does that mean for how p approaches 0 and 1?

Week 3: Deviance, LRT, AIC, confidence regions

Scaled deviance $D = 2(\ell_{\text{sat}} - \ell_{\text{model}})$. For nested models $D_{\text{small}} - D_{\text{big}} \approx \chi^2$ with df = difference in number of parameters. AIC $= -2\ell + 2p$. LR confidence region $= \{\theta : 2(\ell(\hat{\theta}) - \ell(\theta)) \le \chi^2_k(0.95)\}$.
Useful quantiles: $\chi^2_1(0.95) = 3.84$, $\chi^2_2(0.95) = 5.99$, $\chi^2_{13}(0.95) = 22.36$.

Problem 3.1 (11 marks)

Continue with the seed trial of Problem 2.2 (`mod1` output there). Two more models were fitted:

```
> mod0 <- glm(cbind(germ, 50 - germ) ~ temp, family = binomial, data = seeds)
> mod2 <- glm(cbind(germ, 50 - germ) ~ temp * coat, family = binomial, data = seeds)
> c(deviance(mod0), df.residual(mod0), AIC(mod0))
[1]  36.35118  14.00000 101.72754
> summary(mod2)

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept)  -5.74259    0.59863  -9.593   <2e-16 ***
temp          0.29997    0.03074   9.759   <2e-16 ***
coatyes       0.57890    0.84265   0.687    0.492
temp:coatyes  0.01526    0.04521   0.337    0.736
---
(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 294.33  on 15  degrees of freedom
Residual deviance: 11.69  on 12  degrees of freedom
AIC: 81.07
```

(a) Use a likelihood ratio test to decide whether `coat` is needed given `temp`. State the hypotheses, the statistic, its null distribution and your conclusion. Compare the statistic with the square of the Wald z value for `coatyes` in `mod1`.

(b) Test whether the effect of temperature differs between coated and uncoated seeds.

(c) Does `mod1` fit the data adequately? Justify with the deviance, and say why this check is valid here but would not be for ungrouped 0/1 data.

(d) From the AIC, recover the maximised log-likelihood of `mod1`, then the log-likelihood of the saturated model.

(e) Which of the three models does AIC prefer? Why does `mod2` have a larger AIC than `mod1` even though its deviance is smaller?

(f) In `mod2` the `coatyes` estimate changed from 0.857 to 0.579 and lost significance. Explain why in two sentences.

Problem 3.2 (6 marks)

Return to the sample of Problem 1.2 (density $\theta x^{\theta-1}$, $n = 20$, $\sum \log x_i = -12.5$, $\hat\theta = 1.6$, Wald 95% CI $(0.899, 2.301)$).

(a) Write the likelihood ratio statistic $2\big(\ell(\hat\theta) - \ell(\theta_0)\big)$ as a function of θ_0.

(b) Is $\theta_0 = 0.95$ inside the 95% LR confidence interval? Is $\theta_0 = 2.35$?

(c) Compare with the Wald interval and explain why the two intervals differ.

Problem 3.3 (4 marks)

In Week 3's tutorial you drew a 95% confidence region for (α, β) using the log-likelihood ratio.

(a) Which contour of $2\big(\ell(\hat\alpha, \hat\beta) - \ell(\alpha, \beta)\big)$ bounds the region, and why that value?

(b) Why is this region not exactly an ellipse, while the Wald region is?

(c) A logistic fit reports a coefficient 9.2 with SE 40.1 ($z = 0.23$), yet dropping that term raises the deviance by 15.6. Which test do you trust and why?

Week 4: Exponential families, IWLS, Poisson regression

The course form is

$$f(y; \theta, \phi) = \exp\left\{ \frac{y\theta - b(\theta)}{a(\phi)} + c(y, \phi) \right\},$$

with $\mathrm{E}(Y) = b'(\theta)$, $\mathrm{Var}(Y) = a(\phi) b''(\theta)$, canonical link $g = (b')^{-1}$ and variance function $v(\mu) = b''(\theta)$ written in terms of μ. Exam questions ask you to identify θ, ϕ, a, b, c, then the canonical link and $v(\mu)$, showing work.

Problem 4.1 (9 marks)

The Inverse-Gaussian distribution with mean $\mu > 0$ and shape $\lambda > 0$ has density

$$f(y; \mu, \lambda) = \sqrt{\frac{\lambda}{2\pi y^3}} \exp\left(-\frac{\lambda (y - \mu)^2}{2\mu^2 y} \right), \qquad y > 0.$$

(a) Show that it is an exponential family, identifying θ, ϕ, $a(\phi)$, $b(\theta)$ and $c(y, \phi)$.

(b) Verify that $b'(\theta) = \mu$.

(c) Find the canonical link and the variance function $v(\mu)$.

(d) Compared with the Poisson and gamma, how fast does the variance grow with the mean?

Problem 4.2 (5 marks)

Starting from $\int f(y; \theta, \phi)\, dy = 1$, prove that $\mathrm{E}(Y) = b'(\theta)$ and $\mathrm{Var}(Y) = a(\phi) b''(\theta)$. (Hint: differentiate the identity with respect to θ twice.)

Problem 4.3 (7 marks)

IWLS updates β by weighted least squares of the working response $z_i = \eta_i + (y_i - \mu_i) g'(\mu_i)$ on x_i with weights $w_i = 1/\big(g'(\mu_i)^2 v(\mu_i)\big)$.

(a) Write z_i and w_i for a Poisson GLM with log link.

(b) For an intercept-only Poisson model with data $y = (2, 3, 7)$, start at $\beta^{(0)} = 0$ and carry out two IWLS iterations by hand. Compare with the exact MLE.

(c) Explain why IWLS gives the same $\hat\beta$ whatever the value of ϕ, and why this matters for quasi-likelihood.

Problem 4.4 (10 marks)

A city recorded bicycle crash reports on 60 days, with daily rainfall `rain` (mm, 0–38) and `weekend` (1 for Saturday/Sunday).

```
> mod2 <- glm(crashes ~ rain + weekend, family = poisson, data = bike)
> summary(mod2)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  1.21272    0.08764  13.838  < 2e-16 ***
rain         0.05098    0.00619   8.237  < 2e-16 ***
weekend     -0.38352    0.16466  -2.329   0.0199 *
---
(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 133.97  on 59  degrees of freedom
Residual deviance: 75.84  on 57  degrees of freedom
AIC: 261.41

> anova(mod2, test = "Chisq")
        Df Deviance Resid. Df Resid. Dev  Pr(>Chi)
NULL                       59    133.975
rain     1    [A]          58     81.686     [B]
weekend  1     5.845       57      [C]     0.01562 *
> sum(residuals(mod2, type = "pearson")^2)
[1] 68.38325
```

(a) Fill in [A], [B] (just say how small) and [C].

(b) Interpret the coefficients of `rain` and `weekend` on the response scale.

(c) The linear predictor for a weekend day with 10 mm of rain has standard error 0.1506. Estimate the expected number of crashes on such a day and give a 95% CI.

(d) Assess goodness of fit using the residual deviance ($\chi^2_{57}(0.95) = 75.62$). What does the Pearson statistic suggest?

(e) Find the maximised log-likelihood of `mod2`.

Week 5: Gamma GLMs, dispersion, F tests, offsets, diagnostics

When ϕ is unknown: $\hat{\phi} = X^2/(n-p)$ with X^2 the Pearson statistic. Nested models are compared with $F = \dfrac{(D_{\text{small}} - D_{\text{big}})/(\text{df}_{\text{small}} - \text{df}_{\text{big}})}{\hat{\phi}_{\text{big}}}$ on $(\text{df}_{\text{small}} - \text{df}_{\text{big}},\ \text{df}_{\text{big}})$ degrees of freedom. A rate model adds `offset(log t)` so that $\log \mu_i = \log t_i + x_i^T \beta$.

Problem 5.1 (10 marks)

Repair times (`hours`) for 40 machine breakdowns are modelled on machine `age` (years) and `shift` (`day`, `night`).

```
> modA <- glm(hours ~ age + shift, family = Gamma(link = "log"), data = repairs)
> summary(modA)

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.46030    0.18478   2.491   0.0173 *
age          0.10078    0.01879   5.362 4.58e-06 ***
```

```
shiftnight   0.35923    0.15040   2.388   0.0221 *
---
(Dispersion parameter for Gamma family taken to be 0.22332)

    Null deviance: 16.86  on 39  degrees of freedom
Residual deviance: 9.49  on 37  degrees of freedom
AIC: 168.63

> sum(residuals(modA, type = "pearson")^2)
[1] 8.262830
> modB <- glm(hours ~ age, family = Gamma(link = "log"), data = repairs)
> deviance(modB)
[1] 10.77461
> anova(modB, modA, test = "F")
```

(a) Show how the dispersion 0.22332 was obtained. What does it imply about the coefficient of variation of repair times?

(b) Compute the F statistic that `anova(modB, modA, test = "F")` reports and give its degrees of freedom. Conclude at the 5% level ($F_{1,37}(0.95) = 4.11$).

(c) Why is an F test used here rather than comparing the change in deviance with χ^2_1?

(d) Why are the test statistics in the summary labelled `t value` rather than `z value`?

(e) Interpret the `age` and `shiftnight` coefficients.

(f) The canonical link for the gamma is the inverse link. Give one reason to prefer the log link here.

Problem 5.2 (8 marks)

Hospital-acquired infections were counted on 24 wards over different numbers of patient-days (`days`, 302–2997). Predictors: `ward` (`general`, `ICU`) and `handhyg`, the % compliance with hand hygiene.

```
> rmod <- glm(infections ~ offset(log(days)) + ward + handhyg,
+             family = poisson, data = wards)
> summary(rmod)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept) -6.18828    0.66753  -9.270  < 2e-16 ***
wardICU      1.01394    0.32182   3.151  0.00163 **
handhyg     -0.01262    0.01053  -1.198  0.23100
---
    Null deviance: 25.27  on 23  degrees of freedom
Residual deviance: 14.85  on 21  degrees of freedom

> free <- glm(infections ~ ward + handhyg + log(days), family = poisson, data = wards)
> summary(free)$coefficients["log(days)", ]
  Estimate Std. Error    z value   Pr(>|z|)
   1.03879    0.28549    3.63854    0.00027
```

(a) Write the model as an equation for the infection rate per patient-day and explain why the offset is needed.

(b) Interpret `wardICU`.

(c) Estimate the expected number of infections on an ICU ward with 60% compliance over 2000 patient-days. The SE of $x^T\hat{\beta}$ (offset excluded) is 0.2358; give a 95% CI.

(d) Use `free` to test whether the offset assumption (coefficient of log(days) equal to 1) is reasonable.

Problem 5.3 (6 marks, diagnostics)

For `rmod`, ward 20 (general, 40% compliance, 2997 patient-days) had 6 infections, fitted value $\hat{\mu} = 3.714$ and leverage $h = 0.340$. Here $n = 24$ and $p = 3$.

(a) Compute its Pearson residual and studentised Pearson residual.

(b) Is its leverage unusually high? Why might this ward have high leverage?

(c) Cook's distance can be computed as $r_{SP}^2\, h/\big(p(1-h)\big)$. Compute it and comment.

(d) What is a half-normal plot used for, and what pattern signals a problem?

Week 6: Overdispersion, quasi-likelihood, mixture models

Quasi-Poisson/binomial: same $\hat{\beta}$ as Poisson/binomial, $\mathrm{SE}_{\text{quasi}} = \mathrm{SE}_{\text{Poisson}} \times \sqrt{\hat{\phi}}$, t tests, F tests for nested models, no AIC.
Mixture: $P(Z_i = k \mid x_i) = \pi_k f_k(x_i) \big/ \sum_j \pi_j f_j(x_i)$.

Problem 6.1 (12 marks)

Ticks were counted on 50 sheep, with predictors `age` (years), `pasture` (A, B, C) and `wool` (cm).

```
> qbig <- glm(ticks ~ age + pasture + wool, family = quasipoisson, data = sheep)
> summary(qbig)

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.93189    0.36361   2.563  0.01380 *
age          0.22273    0.04352   5.118 6.20e-06 ***
pastureB     0.72865    0.19616   3.715  0.00056 ***
pastureC     0.13993    0.21583   0.648  0.52000
wool         0.06813    0.07907   0.862  0.39300
---
(Dispersion parameter for quasipoisson family taken to be 4.33019)

    Null deviance: 394.77  on 49  degrees of freedom
Residual deviance: 198.10  on 45  degrees of freedom
AIC: NA

> psmall <- glm(ticks ~ age, family = poisson, data = sheep)
> summary(psmall)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  1.58854    0.11988  13.251   <2e-16 ***
age          0.21119    0.02073  10.187   <2e-16 ***
---
(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 394.77  on 49  degrees of freedom
Residual deviance: 283.48  on 48  degrees of freedom
AIC: 498.75

> sum(residuals(psmall, type = "pearson")^2)/48
[1] 5.844822
```

(a) Is there evidence of overdispersion relative to the Poisson? Use two pieces of the output.

(b) `pbig` is the Poisson fit with the same formula as `qbig`. Without refitting, give its estimate and standard error for `pastureB` and for `wool`.

(c) Using `pbig`, is `wool` significant at the 10% level? Using `qbig`? Explain the difference.

(d) Test whether `pasture` and `wool` can be dropped together, allowing for overdispersion. Give the statistic, its degrees of freedom and the conclusion ($F_{3,45}(0.95) = 2.81$). Which of the two dispersion estimates (4.330 or 5.845) belongs in the statistic, and why?

(e) Why is the AIC `NA` for `qbig`?

(f) Give two biological reasons ticks per sheep could be overdispersed.

(g) Summarise in two or three sentences what goes wrong if the overdispersion is ignored.

Problem 6.2 (6 marks)

Suppose $\mathrm{E}(Y) = \mu$ and $\mathrm{Var}(Y) = \phi v(\mu)$, and define the quasi-score $U = (Y - \mu)/\big(\phi v(\mu)\big)$.

(a) Show that $\mathrm{E}(U) = 0$ and $\mathrm{Var}(U) = -\mathrm{E}(\partial U/\partial \mu) = 1/\big(\phi v(\mu)\big)$, the same properties as a true score.

(b) The quasi-log-likelihood is $Q(\mu; y) = \int_y^\mu \frac{y-t}{\phi v(t)}\, dt$. Find Q for $v(\mu) = \mu$ and for $v(\mu) = \mu^2$, and say which ordinary log-likelihoods they match up to $1/\phi$.

(c) Why is a quasi-likelihood estimator usually less efficient than the MLE?

Problem 6.3 (6 marks)

Counts follow a two-component Poisson mixture: $Z_i \sim \mathrm{Categorical}(0.6, 0.4)$, $X_i \mid Z_i = 1 \sim \mathrm{Poisson}(1.5)$, $X_i \mid Z_i = 2 \sim \mathrm{Poisson}(6)$.

(a) Write the marginal pmf of X_i and the (incomplete) log-likelihood of a sample $x_1, \ldots, x_n$. Why can't the MLE be found in closed form?

(b) Compute $P(Z_i = 1 \mid X_i = 3)$. Which cluster is $x = 3$ assigned to by the rule $P(Z_i = 1 \mid x_i) > 0.5$?

(c) Show that the rule assigns x to cluster 1 exactly when $x < c$, and find c.

(d) Name one way a Gaussian mixture likelihood can misbehave that a Poisson mixture cannot.

Week 7: The EM algorithm

The exam question (10–12 marks) is always: write the complete-data log-likelihood with indicators $I(Z_i = k)$, take the expectation over $Z \mid X, \theta^0$ to get $Q(\theta, \theta^0)$ (each indicator becomes $\gamma_{ik} = P(Z_i = k \mid X_i, \theta^0)$), then maximise Q. Expect a twist. You may leave γ_{ik} unexpanded unless asked for the E-step.

Problem 7.1 (12 marks, zero-inflated Poisson)

A survey counts fish caught per angler. Some anglers fished in a spot with no fish and must catch 0. Model, for $i = 1, \ldots, n$:

- $Z_i \sim \mathrm{Categorical}(\pi, 1 - \pi)$;
- $X_i \mid Z_i = 1$ is 0 with probability 1 (a point mass at zero);
- $X_i \mid Z_i = 2 \sim \mathrm{Poisson}(\lambda)$.

Parameters $\theta = (\pi, \lambda)$.

(a) Write $P(X_i = x \mid Z_i = 1)$ and the complete-data log-likelihood $\log P(X, Z \mid \theta)$. Take care with observations where $x_i > 0$.

(b) Derive $Q(\theta, \theta^0)$.

(c) E-step: find $\gamma_i = P(Z_i = 1 \mid X_i = x_i, \theta^0)$ for $x_i = 0$ and for $x_i > 0$.

(d) M-step: derive the updates for π and λ.

(e) The data are $x = (0, 0, 0, 0, 1, 2, 3, 0, 2, 4)$. Starting from $\pi^0 = 0.5$, $\lambda^0 = 2$, carry out one full EM iteration.

Problem 7.2 (8 marks, a shared parameter)

For $i = 1, \ldots, n$: $Z_i \sim$ Categorical$(p, 1-p)$, $X_i \mid Z_i = 1 \sim \mathrm{N}(\mu, 1)$, $X_i \mid Z_i = 2 \sim \mathrm{N}(-\mu, 1)$. Parameters $\theta = (p, \mu)$.

(a) Derive $Q(\theta, \theta^0)$.

(b) Derive the M-step for p and μ.

(c) Interpret the μ update in one sentence.

Problem 7.3 (6 marks, three components)

In a $K = 3$ component mixture with weights (π_1, π_2, π_3), Q contains the term $\sum_i \sum_k \gamma_{ik} \log \pi_k$.

(a) Use a Lagrange multiplier for the constraint $\pi_1 + \pi_2 + \pi_3 = 1$ to show that $\pi_k = \frac{1}{n} \sum_i \gamma_{ik}$.

(b) Why can't you set $\partial Q / \partial \pi_k = 0$ separately for each k without the constraint?

Problem 7.4 (5 marks, behaviour of EM)

(a) Using Jensen's inequality, explain why the incomplete log-likelihood $\log p(x \mid \theta)$ never decreases from one EM iteration to the next.

(b) EM was run from three starting values and returned incomplete log-likelihoods -412.7, -405.3 and -405.3 (the first run at a different parameter value from the other two). Which estimate do you report and why? Why does EM need several starts?

(c) A plot of the incomplete log-likelihood against iteration number goes down at iteration 6. What does that tell you?

Week 8: Bayesian inference and simulating random variables

Posterior $\propto$ prior $\times$ likelihood: drop every factor that does not involve the parameter, then recognise the kernel. Posterior predictive $p(\tilde{x} \mid x) = \int p(\tilde{x} \mid \theta)\, p(\theta \mid x)\, d\theta$. Simulation: discrete via the cdf ladder, continuous via $X = F^{-1}(U)$.

Problem 8.1 (8 marks)

Daily calls to a help line are $X_1, \ldots, X_n$ iid Poisson(θ), with prior $\theta \sim$ Gamma(shape a, rate b), density $b^a \theta^{a-1} e^{-b\theta} / \Gamma(a)$.

(a) Show that the posterior is Gamma$\big(a + \sum x_i,\ b + n\big)$.

(b) Write the posterior mean as a weighted average of the prior mean and the MLE, and say what happens as $n \to \infty$.

(c) With $a = 2$, $b = 1$, $n = 5$ and $\sum x_i = 13$, give the posterior distribution and its mean.

(d) Derive the posterior predictive probability that tomorrow has no calls, and evaluate it.

Problem 8.2 (7 marks)

A lab measures a pollutant with known measurement SD $\sigma = 4$. Four measurements average $\bar{x} = 52$. The prior from neighbouring sites is $\theta \sim \mathrm{N}(45, 6^2)$.

(a) Find the posterior distribution of θ and a 95% credible interval.

(b) Express the prior as $\mathrm{N}(\mu, \sigma^2/n_0)$. What is n_0, and how does it explain the weight on the prior?

(c) A permit is breached if a single future measurement exceeds 60. Find the posterior predictive probability of a breach.

(d) Explain the difference in interpretation between your credible interval and a frequentist 95% CI.

Problem 8.3 (6 marks, reading code)

Consider

```
sim <- function() {
  U <- runif(1)
  y <- 0
  p <- exp(-2)
  F <- p
  while (U > F) {
    y <- y + 1
    p <- p * 2 / y
    F <- F + p
  }
  return(y)
}
```

(a) What distribution does `sim()` return? Justify by saying what `p` and `F` hold after the loop has run k times.

(b) If `runif(1)` returns 0.7, what does `sim()` return? ($e^{-2} = 0.1353$.)

(c) How many times is the condition `U > F` evaluated, on average?

(d) Why is it more efficient to order the values by decreasing probability for a general discrete distribution?

Problem 8.4 (6 marks, inversion)

X has the Laplace density $f(x) = \frac{1}{2}e^{-|x|}$, $x \in \mathbb{R}$.

(a) Find the cdf $F(x)$ and its inverse $F^{-1}(u)$.

(b) Give an algorithm that turns $U \sim U(0,1)$ into X. Apply it to $u = 0.2$ and $u = 0.9$.

(c) Prove that $X = F^{-1}(U)$ has cdf F for any continuous, strictly increasing F.

Week 9: Rejection sampling and Monte Carlo

General rejection: choose an envelope density h you can sample and $k \geq \sup_x f(x)/h(x)$; draw $X \sim h$ and $Y \sim U\big(0, kh(X)\big)$; accept X if $Y < f(X)$. $P(\text{accept}) = 1/k$ for a normalised f ($\int f^*/k$ for an unnormalised f^*), so the number of tries is Geometric with mean k. You may assume generators for the standard distributions unless told otherwise.

Problem 9.1 (8 marks)

Target: $f(x) = 20x(1-x)^3$, $0 < x < 1$ (the Beta$(2,4)$ density).

(a) Using a $U(0,1)$ envelope, find the smallest valid k and the acceptance probability.

(b) Two candidate points are generated: $(X, U_2) = (0.6, 0.5)$ and $(0.3, 0.5)$, where the vertical coordinate is $Y = U_2 \times k \times h(X)$. Which are accepted?

(c) Now use the envelope $h(x) = 3(1-x)^2$ (the Beta$(1,3)$ density). Find the smallest k. Is this better? Explain how you would simulate from h with a single uniform.

Problem 9.2 (5 marks, unnormalised target)

You want to sample from the density proportional to $f^*(x) = e^{-x^2/2}\big(1 + \sin^2(3x)\big)$, $x \in \mathbb{R}$, without computing its normalising constant.

(a) Using a N$(0,1)$ envelope h, find k with $f^*(x) \le k\,h(x)$ for all x.

(b) Write the algorithm.

(c) Given that $\int e^{-x^2/2}\sin^2(3x)\,dx \approx \sqrt{2\pi}/2$, find the acceptance probability and the expected number of normal draws per accepted sample.

Problem 9.3 (6 marks)

Simulate $X \sim$ Gamma(shape 2, rate 1), $f(x) = xe^{-x}$, using an Exponential(μ) envelope $h(x) = \mu e^{-\mu x}$, $0 < \mu < 1$.

(a) Show that $k^*(\mu) = \sup f/h = e^{-1}/\big(\mu(1-\mu)\big)$.

(b) Find the μ that minimises k^* and the resulting acceptance probability.

(c) Why must $\mu < 1$?

Problem 9.4 (6 marks, Monte Carlo for marginals)

Insect eggs: $X \mid N, p \sim \text{Bin}(N,p)$, $N \sim \text{Poisson}(20)$, $p \sim \text{Beta}(3,3)$, with N and p independent.

(a) Write an algorithm to estimate $P(X \ge 12)$ by simulation, using only standard generators.

(b) Find $\text{E}(X)$ exactly and say how you would check your simulation against it.

(c) With 10,000 draws the estimate is 0.36. Give its Monte Carlo standard error.

(d) Write an algorithm to sample from the mixture density $f(x) = 0.3 \times \text{Exp}(1) + 0.7 \times \text{N}(5,1)$ (the composition method).

Week 10: Metropolis–Hastings

The acceptance probability for a move from the current θ_c to the proposed θ_n is

$$A = \min\left\{1,\ \frac{\pi(\theta_n)\,q(\theta_n \to \theta_c)}{\pi(\theta_c)\,q(\theta_c \to \theta_n)}\right\},$$

where π is the posterior up to a constant (prior $\times$ likelihood) and $q(a \to b)$ is the proposal density of b given a. The most common lost marks: forgetting the proposal ratio when q is not symmetric, and evaluating q in the wrong direction.

Problem 10.1 (7 marks)

Counts $y = (2, 4, 3)$ are iid Poisson(λ) with prior $\lambda \sim$ Exponential(rate 1). An MH sampler proposes $\lambda_n \sim$ Gamma(shape 4, rate $4/\lambda_c$), a proposal with mean λ_c and density $(4/\lambda_c)^4 \lambda_n^3 e^{-4\lambda_n/\lambda_c}/\Gamma(4)$.

(a) Write the posterior up to a constant.

(b) Is this proposal symmetric? Explain.

(c) Compute the acceptance probability for $\lambda_c = 2$, $\lambda_n = 4$. Show the posterior ratio and the proposal ratio separately.

Problem 10.2 (9 marks)

One observation $x = 1$ with $x \sim \mathrm{N}(\mu, 1/\tau)$. Independent priors: $\mu \sim \mathrm{N}(0, 4)$ (variance 4) and $\tau \sim$ Gamma(shape 3, rate 2). From the current (μ_c, τ_c), propose $\tau_n \sim$ Gamma(shape 2, rate $2/\tau_c$) and then $\mu_n \sim \mathrm{N}(\mu_c, 1)$.

(a) Write $p(\mu, \tau \mid x)$ up to a constant.

(b) Which part of the proposal ratio cancels and why?

(c) Compute the acceptance probability for $(\mu_c, \tau_c) = (1, 1)$ and $(\mu_n, \tau_n) = (2, 2)$.

Problem 10.3 (8 marks, reading MCMC output)

A random-walk MH sampler with proposal $\mathrm{N}(\theta_c, \sigma^2)$ was run for $n = 5000$ iterations on the same target with three values of σ:

Proposal SD σ	Acceptance rate	Lag-1 autocorrelation ρ
0.1	0.94	0.98
2.5	0.38	0.72
25	0.03	0.95

(a) Why does the proposal ratio not appear for this sampler?

(b) Compute the effective sample size $n(1-\rho)/(1+\rho)$ for each run. Which σ would you use?

(c) Explain in a few sentences why both a very small and a very large σ give high autocorrelation, and what each trace plot would look like.

(d) A symmetric-proposal MH sampler accepts with probability $\min\{1, \exp(-|\theta_n|^3)/\exp(-|\theta_c|^3)\}$. What is its stationary distribution (up to a constant)?

(e) Why does MH only need the posterior up to a normalising constant?

Week 11: Gibbs sampling and MCMC diagnostics

Full conditional: write the joint $p(\text{all parameters}, \text{data})$, keep only the factors containing the parameter of interest, and recognise the kernel (Normal: exp of a quadratic in θ; Gamma: $\theta^{a-1}e^{-b\theta}$). Complete the square for the normal: $\exp\{-(A\theta^2 - 2B\theta)/2\}$ is the $\mathrm{N}(B/A,\ 1/A)$ kernel.

Problem 11.1 (10 marks)

$y_1, \ldots, y_n$ iid $\mathrm{N}(\mu, 1/\tau)$. Independent priors: $\mu \sim \mathrm{N}(m_0, 1/\kappa_0)$ and $\tau \sim$ Gamma(shape a, rate b).

(a) Derive $p(\mu \mid \tau, y)$ and identify it.

(b) Derive $p(\tau \mid \mu, y)$ and identify it.

(c) Write the Gibbs sampler, including initialisation, burn-in and how you would estimate $P(\mu > 2 \mid y)$.

(d) With $n = 4$, $\sum y_i = 10$, $\sum y_i^2 = 30$, $m_0 = 0$, $\kappa_0 = 1$, $a = b = 1$: give the conditional for μ when $\tau = 2$, then the conditional for τ when $\mu = 2.5$.

(e) Why can't you sample directly from $p(\mu, \tau \mid y)$ here, while the normal–gamma prior $\mu \mid \tau \sim \mathrm{N}\big(m_0, 1/(\kappa_0 \tau)\big)$ would allow it?

Problem 11.2 (6 marks, a discrete full conditional)

Yearly accident counts $y = (3, 4, 1, 0, 1)$ follow Poisson(λ_1) up to year m and Poisson(λ_2) after, with change point $m \in \{1, 2, 3, 4\}$ given a uniform prior.

(a) Show that $p(m \mid \lambda_1, \lambda_2, y) \propto (\lambda_1/\lambda_2)^{S_m} e^{-m(\lambda_1 - \lambda_2)}$, where $S_m = y_1 + \cdots + y_m$.

(b) Compute the full conditional of m when $\lambda_1 = 3$, $\lambda_2 = 0.8$.

(c) This conditional is not a named distribution. How does the Gibbs sampler draw from it?

Problem 11.3 (4 marks)

For a bivariate normal with unit variances and correlation ρ, $X_1 \mid X_2 = x_2 \sim \mathrm{N}(\rho x_2, 1 - \rho^2)$. Explain why Gibbs sampling is slow when $\rho = 0.95$, and suggest a remedy.

Problem 11.4 (7 marks, diagnostics)

Three chains of 6000 iterations were started from widely spread values. After discarding the first 1000 of each:

Parameter	SD of pooled 15,000 draws	Mean of the 3 within-chain SDs	Lag-1 autocorrelation
α	1.31	1.02	0.90
β	1.03	1.01	0.60

(a) Compute the (simplified) BGR statistic $\hat{R} = B/W$ for each parameter. Has each converged?

(b) Why must the starting values be overdispersed for this check to be meaningful?

(c) Compute the ESS for β from the pooled draws. Is it enough for the course's ESS > 4000 guideline?

(d) What is burn-in and why do we discard it? What does thinning gain and lose?

(e) Describe in words what the trace plots of α for the three chains probably look like.

Week 12: Variational inference (CAVI)

Mean-field CAVI: $q_j^*(\theta_j) \propto \exp\{\mathrm{E}_{-j}[\log p(\theta, y)]\}$, the expectation taken over the other factors. Method: write $\log p(\theta, y)$, keep the terms with θ_j, replace every other parameter by its q-expectation (using $\mathrm{E}[\mu^2] = \mathrm{V}[\mu] + \mathrm{E}[\mu]^2$), and recognise the kernel.

Problem 12.1 (10 marks)

Use the model of Problem 11.1 with $m_0 = 0$: y_i iid $\mathrm{N}(\mu, 1/\tau)$, $\mu \sim \mathrm{N}(0, 1/\kappa_0)$, $\tau \sim \mathrm{Gamma}(a, b)$ independently, and $q(\mu, \tau) = q_\mu(\mu) q_\tau(\tau)$.

(a) Derive $q_\mu^*(\mu)$ and identify it. Its parameters may involve $\mathrm{E}_\tau[\tau]$.

(b) Derive $q_\tau^*(\tau)$ and identify it. Its parameters may involve $\mathrm{E}_\mu[\mu]$ and $\mathrm{V}_\mu[\mu]$.

(c) With $n = 4$, $\sum y_i = 10$, $\sum y_i^2 = 30$, $\kappa_0 = 1$, $a = b = 1$, start from $\mathrm{E}_\tau[\tau] = 1$ and carry out one full CAVI cycle (update q_μ, then q_τ).

(d) Compare q_μ^* with the Gibbs conditional $p(\mu \mid \tau, y)$ from Problem 11.1. What is the structural difference?

Problem 12.2 (8 marks, a hierarchical Poisson)

One count $y = 4$ is observed, with

$$y \mid \lambda \sim \text{Poisson}(\lambda), \qquad \lambda \mid \beta \sim \text{Gamma}(\text{shape } 2,\ \text{rate } \beta), \qquad \beta \sim \text{Gamma}(\text{shape } 1,\ \text{rate } 1).$$

Approximate $p(\lambda, \beta \mid y)$ by $q_\lambda(\lambda) q_\beta(\beta)$.

(a) Write $\log p(y, \lambda, \beta)$ up to a constant.

(b) Show that $q_\lambda^* = \text{Gamma}\big(y + 2,\ 1 + \mathrm{E}[\beta]\big)$ and $q_\beta^* = \text{Gamma}\big(3,\ 1 + \mathrm{E}[\lambda]\big)$.

(c) Solve for the fixed point of the CAVI updates in closed form and give both variational distributions.

(d) Is q_λ^* the exact marginal posterior of λ? Explain.

Problem 12.3 (6 marks, concepts)

(a) Show that $\log p(y) = \text{ELBO}(q) + \mathrm{KL}\big(q(\theta) \,\|\, p(\theta \mid y)\big)$, and explain why maximising the ELBO is the same as minimising the KL divergence, even though $p(\theta \mid y)$ is unknown.

(b) Why is it called the evidence lower bound?

(c) Mean-field VI tends to underestimate posterior variances. Explain why in terms of $\mathrm{KL}(q \,\|\, p)$, and give one situation where you would still prefer VI to MCMC.

(d) Is the mean-field factorisation $q(\mu, \tau) = q_\mu(\mu) q_\tau(\tau)$ an assumption about the model? Explain.

Part II Solutions

Worked solutions with the steps a marker looks for. Try each question under time first.

Week 1 solutions

The recipe every time: write $\ell(\theta)$, differentiate, set $\ell'(\theta)=0$, then take $-\mathrm{E}[\ell''(\theta)]$. If ℓ'' still contains the data, replace each data sum by its expectation.

1.1

$$\ell(\theta)=\sum_i \log x_i - n\log\theta - \frac{\sum_i x_i^2}{2\theta}, \qquad \ell'(\theta)=-\frac{n}{\theta}+\frac{\sum_i x_i^2}{2\theta^2}, \qquad \ell''(\theta)=\frac{n}{\theta^2}-\frac{\sum_i x_i^2}{\theta^3}.$$

- MLE: $\ell'(\theta)=0$ gives $\hat\theta=\sum x_i^2/(2n)$. Check $\ell''(\hat\theta)=-n/\hat\theta^2<0$, so it is a maximum.
- Fisher information: $\mathrm{E}\big(\sum X_i^2\big)=2n\theta$, so $\mathrm{E}[\ell'']=n/\theta^2-2n/\theta^2=-n/\theta^2$ and $I(\theta)=n/\theta^2$.
- Asymptotics: $\hat\theta \approx \mathrm{N}(\theta,\theta^2/n)$.
- (d) At $\hat\theta$, $\sum x_i^2=2n\hat\theta$, so $-\ell''(\hat\theta)=-n/\hat\theta^2+2n/\hat\theta^2=n/\hat\theta^2=I(\hat\theta)$. Plugging in is fine because $\hat\theta$ is consistent and $I(\theta)/n$ is continuous in θ, so $I(\hat\theta)/I(\theta)\to 1$ and $(\hat\theta-\theta)\sqrt{I(\hat\theta)}$ is still asymptotically $\mathrm{N}(0,1)$ (Slutsky).

1.2

- $\ell(\theta)=n\log\theta+(\theta-1)\sum\log x_i$, $\ell'(\theta)=n/\theta+\sum\log x_i$, so $\hat\theta=-n/\sum\log x_i$.
- $\ell''(\theta)=-n/\theta^2$ contains no data, so $I(\theta)=n/\theta^2$ and $\hat\theta\approx\mathrm{N}(\theta,\theta^2/n)$.
- Numbers: $\hat\theta=20/12.5=1.6$. SE $=\hat\theta/\sqrt{n}=1.6/\sqrt{20}=0.358$. 95% CI: $1.6\pm1.96\times0.358=(0.899,\ 2.301)$.
- Invariance: the MLE of $\theta/(\theta+1)$ is $1.6/2.6=0.615$.

1.3

$$\ell(m)=n\log\tfrac{27}{2}-3n\log m+2\sum_i\log x_i-\frac{3\sum_i x_i}{m}, \qquad \ell'(m)=-\frac{3n}{m}+\frac{3\sum_i x_i}{m^2}.$$

- $\ell'(m)=0$ gives $\hat m=\bar x$.
- $\ell''(m)=3n/m^2-6\sum x_i/m^3$ depends on $\sum x_i$. Using $\mathrm{E}\big(\sum X_i\big)=nm$: $\mathrm{E}[\ell'']=3n/m^2-6n/m^2=-3n/m^2$, so $I(m)=3n/m^2$.
- $\hat m\approx\mathrm{N}\big(m,\ m^2/(3n)\big)$. Exact check: $\mathrm{Var}(X)=\text{shape}\times\text{scale}^2=3(m/3)^2=m^2/3$, so $\mathrm{Var}(\bar X)=m^2/(3n)$ exactly. The asymptotic variance is exact here.
- Marker's point: a common error is to stop at $-\ell''(m)$ with $\sum x_i$ left in it. That is the *observed* information, not $I(m)$.

1.4

Write $S = \sum (x_i - 1)^2 / x_i$.

$$\ell(\lambda) = \frac{n}{2}\log\lambda - \frac{n}{2}\log(2\pi) - \frac{3}{2}\sum_i \log x_i - \frac{\lambda S}{2}, \qquad \ell'(\lambda) = \frac{n}{2\lambda} - \frac{S}{2}, \qquad \ell''(\lambda) = -\frac{n}{2\lambda^2}.$$

- $\hat\lambda = n/S$, $I(\lambda) = n/(2\lambda^2)$, so $\hat\lambda \approx \mathrm{N}(\lambda,\ 2\lambda^2/n)$.
- Numbers: $\hat\lambda = 50/12.5 = 4$. SE $= \hat\lambda\sqrt{2/n} = 4 \times 0.2 = 0.8$. 95% CI: $4 \pm 1.568 = (2.43,\ 5.57)$.

Week 2 solutions

2.1

$$\ell(\beta) = \sum_i \Big[y_i \log p_i + (m_i - y_i)\log(1 - p_i) \Big] = \sum_i \Big[y_i\, x_i^T\beta - m_i \log\left(1 + e^{x_i^T\beta}\right) \Big].$$

- Score: $\partial/\partial\beta_j$ of $m_i \log(1 + e^{\eta_i})$ is $m_i p_i x_{ij}$, so $\partial\ell/\partial\beta_j = \sum_i x_{ij}(y_i - m_i p_i)$.
- Second derivative: $\partial p_i/\partial\beta_k = p_i(1 - p_i)x_{ik}$, so $\partial^2\ell/\partial\beta_j\partial\beta_k = -\sum_i m_i p_i(1 - p_i)x_{ij}x_{ik}$, i.e. the Hessian is $-X^T W X$.
- The Hessian has no y_i in it, so taking the expectation changes nothing: observed = expected information. This happens because logit is the canonical link.
- $\hat\beta \approx \mathrm{N}\big(\beta, (X^T W X)^{-1}\big)$. $\mathrm{SE}(\hat\beta_j)$ = square root of the j-th diagonal entry of $(X^T \hat W X)^{-1}$, with W evaluated at $\hat p$ (this is the `Std. Error` column).

2.2

(a) Coated seeds have $\exp(0.85723) = 2.36$ times the odds of germinating of uncoated seeds at the same temperature. 95% CI on the log-odds scale: $0.85723 \pm 1.96 \times 0.17635 = (0.512,\ 1.203)$; exponentiate: $(1.67,\ 3.33)$.

(b) Odds multiply by $\exp(5 \times 0.30714) = 4.64$.

(c) $\hat p = 1/(1 + e^{-0.50643}) = 0.624$. CI for η: $0.50643 \pm 1.96 \times 0.12415 = (0.263,\ 0.750)$. Transform with the inverse logit: $(0.565,\ 0.679)$.

(d) The asymptotic normality of $\hat\beta$ carries over linearly to $\hat\eta = x^T\hat\beta$, so the normal approximation is good on the η scale. Transforming its endpoints with the monotone inverse link keeps 95% coverage and keeps the interval inside $(0, 1)$. A symmetric interval on p can spill outside $(0, 1)$ and is poor when p is near 0 or 1.

(e) Solve $\eta = 0$: $-5.87928 + 0.30714 \times \text{temp} + 0.85723 = 0$, so temp $= 5.02205/0.30714 = 16.4\,^\circ$C.

2.3

(a) Logit: $1/(1 + e^{-0.4}) = 0.599$. Probit: $\Phi(0.4) = 0.655$. Cloglog: $1 - \exp(-e^{0.4}) = 1 - \exp(-1.492) = 0.775$.

(b) Cloglog is asymmetric: p approaches 0 slowly but approaches 1 very fast (like a double exponential). Logit and probit are both symmetric about $\eta = 0$, $p = 0.5$; probit has lighter tails, so it reaches 0 and 1 a little faster than logit.

Week 3 solutions

3.1

(a) $H_0 : \beta_{\text{coat}} = 0$ (`mod0`) vs $H_1 : \beta_{\text{coat}} \neq 0$ (`mod1`). LR statistic $= D_0 - D_1 = 36.35 - 11.80 = 24.55$ on $14 - 13 = 1$ df. Under H_0 it is approximately χ^2_1; $24.55 > 3.84$ ($p \approx 7 \times 10^{-7}$), so keep `coat`. Wald: $z^2 = 4.861^2 = 23.63$, close to 24.55: the two tests are asymptotically equivalent and agree here.

(b) LR statistic $= 11.80 - 11.69 = 0.11$ on 1 df, far below 3.84 (Wald $z = 0.337$, $p = 0.74$). No evidence of an interaction: the temperature slope is the same for both coatings, and `mod1` is enough.

(c) $D_1 = 11.80$ on 13 df; under an adequate model $D \approx \chi^2_{13}$ with 95% point 22.36 ($p \approx 0.54$). No lack of fit. Valid because every $m_i = 50$ is large, so the saturated model has a fixed number of parameters per group and the χ^2 approximation holds. With 0/1 data the saturated model gains one parameter per observation, D depends on the data only through $\hat{p}$, and the χ^2 approximation fails.

(d) AIC $= -2\ell + 2p$ with $p = 3$: $\ell(\text{mod1}) = (6 - 79.18)/2 = -36.59$. Since $D = 2(\ell_{\text{sat}} - \ell)$: $\ell_{\text{sat}} = -36.59 + 11.80/2 = -30.69$.

(e) AIC: `mod0` 101.73, `mod1` 79.18, `mod2` 81.07, so `mod1`. Adding the interaction lowered the deviance by only 0.11 (so -2ℓ falls by 0.11) but costs 2 for the extra parameter: net +1.89. AIC rewards fit and penalises complexity; any added term must lower the deviance by more than 2.

(f) In `mod2`, `coatyes` is the coating effect at temp = 0 °C, far outside the 12–26 °C data, so it is a different quantity from the common shift in `mod1`. It is also strongly correlated with the interaction estimate, which inflates its SE from 0.18 to 0.84.

3.2

$$2\big(\ell(\hat{\theta}) - \ell(\theta_0)\big) = 2\Big[n \log \frac{\hat{\theta}}{\theta_0} + (\hat{\theta} - \theta_0) \sum_i \log x_i\Big] = 2\Big[20 \log \frac{1.6}{\theta_0} - 12.5(1.6 - \theta_0)\Big].$$

- $\theta_0 = 0.95$: $2[20 \log(1.6/0.95) - 12.5 \times 0.65] = 2[10.426 - 8.125] = 4.60 > 3.84$, so 0.95 is *outside* the LR interval, though inside the Wald one.
- $\theta_0 = 2.35$: $2[20 \log(1.6/2.35) + 12.5 \times 0.75] = 2[-7.688 + 9.375] = 3.37 < 3.84$, so 2.35 is *inside* the LR interval, though outside the Wald one.
- The LR interval is about (0.997, 2.407), shifted right of the Wald (0.899, 2.301). The log-likelihood is not quadratic: it is skewed (falls steeply towards 0, slowly for large θ). The Wald interval forces symmetry by using a quadratic approximation at $\hat{\theta}$; the LR interval follows the actual likelihood and is usually more accurate in small samples.

3.3

(a) The contour at $\chi^2_2(0.95) = 5.99$, because $2(\ell(\hat{\theta}) - \ell(\theta)) \approx \chi^2$ with 2 df when both parameters are tested at once.

(b) The Wald region uses the quadratic approximation $(\hat{\theta} - \theta)^T I(\hat{\theta})(\hat{\theta} - \theta) \leq 5.99$, which is an ellipse. The LR region uses the true log-likelihood surface, which is only approximately quadratic.

(c) Trust the LRT. A huge SE with a modest deviance drop is the Hauck–Donner effect, typical with (near) separation: as $|\beta|$ grows, the curvature-based SE grows faster, so the Wald statistic shrinks and understates the evidence. A deviance drop of 15.6 on 1 df is highly significant.

Week 4 solutions

4.1

Expand the square: $(y-\mu)^2/(\mu^2 y) = y/\mu^2 - 2/\mu + 1/y$. So

$$\log f = \lambda\Big[y\Big(-\frac{1}{2\mu^2}\Big) + \frac{1}{\mu}\Big] - \frac{\lambda}{2y} + \frac{1}{2}\log\frac{\lambda}{2\pi y^3}.$$

- $\theta = -1/(2\mu^2)$, so $\mu = (-2\theta)^{-1/2}$ and $1/\mu = \sqrt{-2\theta}$. Then $b(\theta) = -1/\mu = -\sqrt{-2\theta}$.
- $\phi = 1/\lambda$, $a(\phi) = \phi$, $c(y,\phi) = -\dfrac{1}{2\phi y} - \dfrac{1}{2}\log(2\pi\phi y^3)$.
- $b'(\theta) = (-2\theta)^{-1/2} = \mu$. ✓
- Canonical link: θ as a function of μ, $g(\mu) = -1/(2\mu^2)$, equivalently $1/\mu^2$ (constants don't matter; R calls it `1/mu^2`).
- $b''(\theta) = (-2\theta)^{-3/2} = \mu^3$, so $v(\mu) = \mu^3$ and $\mathrm{Var}(Y) = \mu^3/\lambda$.
- Poisson $v = \mu$, gamma $v = \mu^2$, inverse Gaussian $v = \mu^3$: the most strongly right-skewed of the three, for responses whose spread grows fastest with the mean.

4.2

Differentiate $\int \exp\{(y\theta - b(\theta))/a(\phi) + c\}\,dy = 1$ with respect to θ:

$$\int \frac{y - b'(\theta)}{a(\phi)} f\,dy = 0 \;\Rightarrow\; \mathrm{E}(Y) = b'(\theta);$$

and differentiating again,

$$\int \Big[\frac{(y-b'(\theta))^2}{a(\phi)^2} - \frac{b''(\theta)}{a(\phi)}\Big] f\,dy = 0 \;\Rightarrow\; \mathrm{Var}(Y) = a(\phi)\,b''(\theta).$$

(Differentiating under the integral is allowed because the support does not depend on θ.)

4.3

(a) $g(\mu) = \log\mu$, $g'(\mu) = 1/\mu$, $v(\mu) = \mu$: $z_i = \eta_i + (y_i - \mu_i)/\mu_i$ and $w_i = 1/\big((1/\mu_i)^2\mu_i\big) = \mu_i$.

(b) Intercept only, equal weights, so the WLS fit is the mean of z.

- Iteration 1: $\beta^{(0)} = 0$, $\mu = 1$, $z = (1, 2, 6)$, $\beta^{(1)} = 3$.
- Iteration 2: $\mu = e^3 = 20.09$, $z_i = 3 + (y_i - 20.09)/20.09$, $\beta^{(2)} = 3 + (4 - 20.09)/20.09 = 2.199$.
- Exact MLE: $\hat\beta = \log\bar y = \log 4 = 1.386$. A bad start overshoots, but the iterations move steadily towards it.

(c) ϕ only scales all the weights by the same constant ($w_i = 1/\big(g'(\mu_i)^2\phi v(\mu_i)\big)$ in general), and WLS estimates do not change when all weights are multiplied by a constant. So $\hat\beta$ needs only g and v, not ϕ or the full distribution, which is exactly what quasi-likelihood specifies.

4.4

(a) $[A] = 133.975 - 81.686 = 52.289$. $[B]$ is the χ^2_1 tail beyond 52.3: about 5×10^{-13}, tiny. $[C] = 81.686 - 5.845 = 75.841$.

(b) Each extra mm of rain multiplies the expected crash count by $e^{0.05098} = 1.052$ (5.2% more), weekend fixed. Weekends have $e^{-0.38352} = 0.68$ times the weekday rate (32% fewer), rain fixed.

(c) $\hat\eta = 1.21272 + 10 \times 0.05098 - 0.38352 = 1.339$, $\hat\mu = e^{1.339} = 3.82$ crashes. CI: $\exp(1.339 \pm 1.96 \times 0.1506) = (2.84,\ 5.13)$.

(d) $D = 75.84$ on 57 df, just above 75.62 ($p \approx 0.048$): borderline evidence of lack of fit at 5%. Pearson $X^2/\text{df} = 68.38/57 = 1.20$, mild overdispersion, so Poisson SEs may be about $\sqrt{1.2} \approx 1.10$ times too small; `weekend` ($p = 0.02$) is the result to check with a quasi-Poisson fit.

(e) $\ell = (2p - \text{AIC})/2 = (6 - 261.41)/2 = -127.70$.

Week 5 solutions

5.1

(a) $\hat{\phi} = X^2/(n-p) = 8.26283/(40-3) = 0.22332$. For the gamma, $\text{Var}(Y) = \phi\mu^2$, so the coefficient of variation is $\sqrt{\phi} \approx 0.47$, constant across machines: the SD of a repair time is about 47% of its mean.

(b) $F = \dfrac{(10.77461 - 9.49145)/(38 - 37)}{0.22332} = \dfrac{1.28316}{0.22332} = 5.75$ on $(1, 37)$ df. $5.75 > 4.11$ ($p \approx 0.022$), so keep `shift`.

(c) The change in deviance is $\phi\times$(change in scaled deviance). Only the scaled change is approximately χ^2, and ϕ is unknown, so we divide by $\hat{\phi}$. Estimating ϕ adds sampling variability, giving a ratio of a χ^2 to an independent χ^2/df: an F distribution. Comparing 1.28 with χ^2_1 would treat ϕ as 1 and be badly wrong.

(d) SEs are scaled by the estimated $\sqrt{\hat{\phi}}$, so Wald statistics are referred to t with $n - p = 37$ df, not $\text{N}(0, 1)$.

(e) With a log link, each extra year of age multiplies the mean repair time by $e^{0.10078} = 1.106$ (about 11% longer), shift fixed. Night-shift repairs take $e^{0.35923} = 1.43$ times as long on average (43% longer), age fixed.

(f) The log link keeps $\mu > 0$ automatically (the inverse link can give negative or infinite fitted means), and it gives multiplicative, easy-to-interpret effects.

5.2

(a) $\log(\mu_i/\text{days}_i) = \beta_0 + \beta_1\,\text{ICU}_i + \beta_2\,\text{handhyg}_i$, i.e. $\mu_i = \text{days}_i \times \exp(x_i^T\beta)$. Wards were observed for different lengths of exposure, so the counts are not comparable directly. The offset log(days) enters with coefficient fixed at 1, so the model describes the rate per patient-day.

(b) ICU wards have $e^{1.01394} = 2.76$ times the infection rate of general wards with the same hand-hygiene compliance.

(c) $\hat{\eta} = \log 2000 + (-6.18828 + 1.01394 - 0.01262 \times 60) = 7.6009 - 5.9315 = 1.669$, $\hat{\mu} = e^{1.669} = 5.31$ infections (2.65 per 1000 patient-days). CI: $\exp(1.669 \pm 1.96 \times 0.2358) = (3.34,\ 8.43)$. The offset is known, so it adds no uncertainty.

(d) H_0: coefficient $= 1$. $z = (1.03879 - 1)/0.28549 = 0.14$, far below 1.96. The data are consistent with counts proportional to exposure, so the offset is justified. (The R p-value tests coefficient $= 0$, which is the wrong null here.)

5.3

(a) $r_P = (6 - 3.714)/\sqrt{3.714} = 2.286/1.927 = 1.19$. With $\phi = 1$: $r_{SP} = 1.19/\sqrt{1 - 0.340} = 1.46$. Not an outlier ($|r| < 2$).

(b) Average leverage $= p/n = 3/24 = 0.125$, so 0.340 is 2.7 times the average (above the $2p/n = 0.25$ rule of thumb). The ward has the lowest compliance (40%, an extreme x value) and the largest exposure, so a large fitted mean and a large IWLS weight.

(c) $D = 1.46^2 \times 0.340/(3 \times 0.660) = 0.37$. Larger than the other wards' but well below 1: high leverage, moderate residual, some influence, nothing alarming. Refit without it to check the `handhyg` estimate.

(d) Plot the ordered absolute residuals, leverages or Cook's distances against half-normal quantiles $\Phi^{-1}\big((n+i)/(2n+1)\big)$. We expect a smooth curve; a point that jumps away from the rest at the top end is a candidate outlier or influential point. A straight line is not expected.

Week 6 solutions

6.1

(a) Yes. The dispersion estimate X^2/df is 4.33 for the big model (5.84 for `psmall`), far above 1. Also the residual deviance 198.10 on 45 df is about 4.4 times its df, whereas under the Poisson it should be about χ^2_{45} (mean 45).

(b) Estimates are identical: 0.72865 and 0.06813. Poisson SE = quasi SE$/\sqrt{\hat\phi}$, with $\sqrt{4.33019} = 2.0809$: pastureB $0.19616/2.0809 = 0.0943$; wool $0.07907/2.0809 = 0.0380$.

(c) Poisson: $z = 0.06813/0.0380 = 1.79$, $p = 0.073 < 0.10$, "significant". Quasi: $t = 0.862$, $p = 0.39$, not. The Poisson SE is too small by the factor $\sqrt{4.33} \approx 2.08$ because it assumes $\text{Var}(Y) = \mu$ when the data vary about 4.3 times more.

(d) Deviances for the two models are the same under Poisson and quasi-Poisson.

$$F = \frac{(283.48 - 198.10)/(48 - 45)}{4.33019} = \frac{28.46}{4.33} = 6.57$$

on $(3, 45)$ df, and $6.57 > 2.81$ ($p \approx 0.0009$): keep pasture/wool (at least one of them). Use $\hat\phi = 4.330$ from the bigger model: it is a valid estimate under both H_0 and H_1, while 5.845 is inflated if the dropped terms matter. Naively comparing 85.4 with χ^2_3 would greatly overstate the evidence.

(e) Quasi-likelihood specifies only the mean and variance, not a distribution, so there is no true likelihood and $\text{AIC} = -2\ell + 2p$ cannot be computed.

(f) Unmeasured sheep-to-sheep heterogeneity in the rate (health, coat condition, behaviour), so the mean varies beyond what the predictors explain; and clustering/contagion, since ticks arrive in groups (larvae from one clutch) so counts are not independent events.

(g) Point estimates are unaffected, but SEs, CIs and p-values are too small and F/χ^2 statistics too large, so you find spurious significance and keep too many predictors. Model-adequacy checks with the deviance also fail.

6.2

(a) $\text{E}(U) = (\text{E}Y - \mu)/(\phi v) = 0$. $\text{Var}(U) = \text{Var}(Y)/(\phi v)^2 = \phi v/(\phi v)^2 = 1/(\phi v)$. $\partial U/\partial\mu = \big[-\phi v - (Y-\mu)\phi v'\big]/(\phi v)^2$; the second term has mean 0, so $-\text{E}(\partial U/\partial\mu) = 1/(\phi v)$.

(b) $v(\mu) = \mu$: $Q = \frac{1}{\phi}\big[y\log t - t\big]_y^\mu = (y\log\mu - \mu - y\log y + y)/\phi$, the Poisson log-likelihood (up to terms free of μ) divided by ϕ.
$v(\mu) = \mu^2$: $Q = \frac{1}{\phi}\big[-y/t - \log t\big]_y^\mu = (-y/\mu - \log\mu + 1 + \log y)/\phi$, the gamma (exponential) log-likelihood kernel.

(c) It uses only the first two moments. When the true distribution is known, the MLE uses all of its information and achieves the Cramér–Rao bound; the quasi estimator is consistent but generally has larger variance.

6.3

(a) $p(x) = 0.6\,e^{-1.5}1.5^x/x! + 0.4\,e^{-6}6^x/x!$, and $\ell(\theta) = \sum_i \log\big[\pi_1 f_1(x_i) + \pi_2 f_2(x_i)\big]$. The log of a sum does not split into separate terms per parameter, so the score equations are coupled and have no closed-form solution. This is why EM is used.

(b) Numerator $0.6 \times e^{-1.5} \times 1.5^3/3! = 0.6 \times 0.22313 \times 0.5625 = 0.07531$. Other term $0.4 \times e^{-6} \times 6^3/3! = 0.4 \times 0.0024788 \times 36 = 0.03569$. $P(Z = 1 \mid X = 3) = 0.07531/0.11100 = 0.678 > 0.5$: cluster 1.

(c) Cluster 1 iff $\pi_1 e^{-\lambda_1}\lambda_1^x > \pi_2 e^{-\lambda_2}\lambda_2^x$ (the $x!$ cancels) iff $x\log(\lambda_2/\lambda_1) < \log(\pi_1/\pi_2) + \lambda_2 - \lambda_1$, i.e.

$$x < c = \frac{\log 1.5 + 4.5}{\log 4} = \frac{4.906}{1.386} = 3.54.$$

So $x \le 3$ goes to cluster 1 and $x \ge 4$ to cluster 2.

(d) With unequal variances, put one mean at a data point and let that $\sigma \to 0$: the likelihood goes to infinity, so the global MLE is degenerate. (Also: label switching means the components are only identifiable up to relabelling, true of both.)

Week 7 solutions

7.1

(a) $P(X_i = x \mid Z_i = 1) = I(x = 0)$. Complete-data log-likelihood:

$$\log P(X, Z \mid \theta) = \sum_{i=1}^{n} \Big\{ I(Z_i = 1)\big[\log\pi + \log I(x_i = 0)\big] + I(Z_i = 2)\big[\log(1-\pi) + x_i\log\lambda - \lambda - \log x_i!\big] \Big\}.$$

If $x_i > 0$ then $Z_i = 2$ for certain, so the $\log I(\cdot) = \log 0$ term never arises with positive probability.

(b) Replace each indicator by its conditional expectation, $\gamma_i = P(Z_i = 1 \mid x_i, \theta^0)$:

$$Q(\theta, \theta^0) = \sum_i \gamma_i \log\pi + \sum_i (1-\gamma_i)\big[\log(1-\pi) + x_i\log\lambda - \lambda - \log x_i!\big].$$

(c) For $x_i = 0$: $\gamma_i = \dfrac{\pi^0}{\pi^0 + (1-\pi^0)e^{-\lambda^0}}$ (Bayes' rule; $P(X = 0 \mid Z = 2) = e^{-\lambda}$). For $x_i > 0$: $\gamma_i = 0$.

(d) $\partial Q/\partial\pi = \sum\gamma_i/\pi - \sum(1-\gamma_i)/(1-\pi) = 0$ gives $\pi = \frac{1}{n}\sum_i \gamma_i$.
$\partial Q/\partial\lambda = \sum(1-\gamma_i)(x_i/\lambda - 1) = 0$ gives $\lambda = \dfrac{\sum(1-\gamma_i)x_i}{\sum(1-\gamma_i)} = \dfrac{\sum x_i}{\sum(1-\gamma_i)}$, since $\gamma_i x_i = 0$ always.

(e) Five zeros, $\sum x_i = 12$. E-step: $\gamma = 0.5/(0.5 + 0.5e^{-2}) = 1/(1 + 0.1353) = 0.8808$ for each zero, 0 otherwise. M-step: $\pi^1 = 5 \times 0.8808/10 = 0.440$; $\lambda^1 = 12/(10 - 4.404) = 12/5.596 = 2.144$. (The incomplete log-likelihood rises from -14.335 to -14.233; EM converges to $\hat\pi = 0.431$, $\hat\lambda = 2.109$.)

7.2

(a) With $\gamma_{i1} = P(Z_i = 1 \mid x_i, \theta^0)$ and $\gamma_{i2} = 1 - \gamma_{i1}$:

$$Q(\theta, \theta^0) = \sum_i \Big\{ \gamma_{i1} \Big[\log p - \tfrac{1}{2}\log 2\pi - \tfrac{(x_i-\mu)^2}{2} \Big] + \gamma_{i2} \Big[\log(1-p) - \tfrac{1}{2}\log 2\pi - \tfrac{(x_i+\mu)^2}{2} \Big] \Big\}.$$

(b) $p = \frac{1}{n}\sum \gamma_{i1}$ (as for any two-component weight). $\partial Q/\partial\mu = \sum \big[\gamma_{i1}(x_i - \mu) - \gamma_{i2}(x_i + \mu)\big] = \sum(\gamma_{i1} - \gamma_{i2})x_i - n\mu = 0$, so $\mu = \frac{1}{n}\sum(\gamma_{i1} - \gamma_{i2})x_i$.

(c) Points believed to come from the $-\mu$ component are sign-flipped, then everything is averaged, so both components inform the one shared mean.

7.3

(a) Let $N_k = \sum_i \gamma_{ik}$. Maximise $\sum_k N_k \log \pi_k - \alpha\big(\sum_k \pi_k - 1\big)$: $N_k/\pi_k = \alpha$, so $\pi_k = N_k/\alpha$. Summing over k: $1 = \sum N_k/\alpha = n/\alpha$ (each row of γ sums to 1), so $\alpha = n$ and $\pi_k = N_k/n$.

(b) Unconstrained, $\partial Q/\partial\pi_k = N_k/\pi_k > 0$ is never zero: Q increases without bound as each π_k grows. The constraint (or writing $\pi_3 = 1 - \pi_1 - \pi_2$) is what produces a maximum.

7.4

(a) For any $q(z)$, by Jensen (log is concave),

$$\log p(x \mid \theta) = \log \sum_z q(z) \frac{p(x, z \mid \theta)}{q(z)} \;\geq\; \sum_z q(z) \log \frac{p(x, z \mid \theta)}{q(z)}.$$

With $q = p(z \mid x, \theta^0)$ the right side is $Q(\theta, \theta^0) + H(q)$, with equality at $\theta = \theta^0$. The M-step picks θ^1 with $Q(\theta^1, \theta^0) \geq Q(\theta^0, \theta^0)$, so $\log p(x \mid \theta^1) \geq Q(\theta^1, \theta^0) + H \geq Q(\theta^0, \theta^0) + H = \log p(x \mid \theta^0)$.

(b) Report the estimate with log-likelihood -405.3, the largest: the MLE is the global maximiser. EM only guarantees convergence to a stationary point (a local maximum here, -412.7), which depends on the start.

(c) A decrease is impossible for a correct EM, so there is a bug: in the E-step, the M-step, or the log-likelihood calculation.

Week 8 solutions

8.1

(a) $p(\theta \mid x) \propto \theta^{a-1}e^{-b\theta} \times \prod_i \theta^{x_i}e^{-\theta} = \theta^{a+\sum x_i - 1}e^{-(b+n)\theta}$, the Gamma$\big(a + \sum x_i,\ b + n\big)$ kernel.

(b) $\dfrac{a + \sum x_i}{b + n} = \dfrac{b}{b+n} \cdot \dfrac{a}{b} + \dfrac{n}{b+n} \cdot \bar{x}$: prior mean a/b with weight $b/(b+n)$, MLE $\bar{x}$ with weight $n/(b+n)$. As $n \to \infty$ the weight on $\bar{x} \to 1$ and the posterior variance $(a + \sum x_i)/(b+n)^2 \to 0$, so the posterior concentrates at the MLE and the prior stops mattering.

(c) Gamma$(15, 6)$, mean $15/6 = 2.5 = \frac{1}{6}(2) + \frac{5}{6}(2.6)$.

(d) Using the Gamma$(15, 6)$ normalising constant:

$$P(\tilde{X} = 0 \mid x) = \int_0^\infty e^{-\theta} \, \frac{6^{15}\theta^{14}e^{-6\theta}}{\Gamma(15)} \, d\theta = \frac{6^{15}}{\Gamma(15)} \cdot \frac{\Gamma(15)}{7^{15}} = \Big(\frac{6}{7}\Big)^{15} = 0.099.$$

(The full predictive is negative binomial. Plugging in $\hat{\theta} = 2.6$ would give $e^{-2.6} = 0.074$, too small, because it ignores uncertainty about θ.)

8.2

(a) Posterior precision $= 1/36 + n/\sigma^2 = 0.0278 + 0.25 = 0.2778$, so variance 3.6. Mean $= 3.6 \times (45/36 + 52 \times 0.25) = 3.6 \times 14.25 = 51.3$. $\theta \mid x \sim \mathrm{N}(51.3,\ 3.6)$; 95% credible interval $51.3 \pm 1.96 \times 1.897 = (47.6,\ 55.0)$.

(b) $36 = 16/n_0$ gives $n_0 = 0.44$: the prior is worth less than half a measurement. The posterior mean is $(n_0\mu + n\bar{x})/(n_0 + n) = (0.44 \times 45 + 4 \times 52)/4.44 = 51.3$, so the data get 90% of the weight. Posterior variance $\sigma^2/(n_0 + n) = 3.6$.

(c) Predictive: $\tilde{X} \mid x \sim \mathrm{N}(51.3,\ 3.6{+}16) = \mathrm{N}(51.3,\ 19.6)$. $P(\tilde{X} > 60) = 1 - \Phi\big((60-51.3)/4.427\big) = 1 - \Phi(1.97) = 0.025$. The predictive variance adds the measurement variance to the posterior variance.

(d) Credible: given the data and prior, θ lies in $(47.6,\ 55.0)$ with probability 0.95, a statement about θ. Frequentist: θ is fixed; 95% of intervals built this way over repeated samples would cover it, and any single interval either does or does not.

8.3

(a) After k passes, $p = e^{-2}2^k/k!$ and $F = \sum_{j \le k} e^{-2}2^j/j!$: the Poisson(2) pmf and cdf. The loop stops at the first y with $F(y) \ge U$, so $P(Y = y) = P\big(F(y-1) < U \le F(y)\big) = p(y)$: $Y \sim \mathrm{Poisson}(2)$.

(b) $F(0) = 0.135$, $F(1) = 0.406$, $F(2) = 0.677$, $F(3) = 0.857$. $U = 0.7$ first falls below F at $y = 3$, so `sim()` returns 3.

(c) The condition is checked $Y + 1$ times, so on average $\mathrm{E}(Y) + 1 = 3$ times.

(d) The expected number of comparisons is $\sum$(position of value) $\times$ (its probability), so checking the most likely values first makes the search stop sooner on average.

8.4

(a) $F(x) = \frac{1}{2}e^x$ for $x < 0$ and $F(x) = 1 - \frac{1}{2}e^{-x}$ for $x \ge 0$. Inverse: $F^{-1}(u) = \log(2u)$ for $u < \frac{1}{2}$ and $-\log\big(2(1-u)\big)$ for $u \ge \frac{1}{2}$.

(b) Draw $U \sim U(0,1)$; if $U < \frac{1}{2}$ return $\log(2U)$, else return $-\log\big(2(1-U)\big)$. $u = 0.2$ gives $\log 0.4 = -0.916$; $u = 0.9$ gives $-\log 0.2 = 1.609$.

(c) $P\big(F^{-1}(U) \le x\big) = P\big(U \le F(x)\big) = F(x)$, using that F is increasing and $P(U \le t) = t$ for $t \in [0,1]$.

Week 9 solutions

9.1

(a) $f'(x) = 20(1-x)^2(1-4x) = 0$ at $x = 1/4$, so $k = f(1/4) = 20 \times \frac{1}{4} \times \frac{27}{64} = 2.109$. $P(\text{accept}) = 1/k = 0.474$: on average 2.1 candidates per sample.

(b) $Y = 0.5 \times 2.109 = 1.055$ in both cases. $f(0.6) = 0.768 < 1.055$: reject. $f(0.3) = 2.058 > 1.055$: accept, return 0.3.

(c) $f/h = \frac{20}{3}x(1-x)$, maximised at $x = \frac{1}{2}$: $k = 5/3 = 1.667$, $P(\text{accept}) = 0.6$. Better, because h follows the shape of f more closely (both are high near 0 and vanish at 1). $H(x) = 1 - (1-x)^3$, so by inversion $X = 1 - (1-U)^{1/3}$, or equivalently $1 - U^{1/3}$.

9.2

(a) $f^*/h = \sqrt{2\pi}\big(1+\sin^2(3x)\big) \le 2\sqrt{2\pi}$, so $k = 2\sqrt{2\pi} \approx 5.01$.

(b) Repeat: draw $X \sim \mathrm{N}(0,1)$ and $U \sim U(0,1)$; if $U < f^*(X)/\big(kh(X)\big) = \big(1+\sin^2(3X)\big)/2$, return X.

(c) $P(\text{accept}) = \int f^*/k = \dfrac{\sqrt{2\pi}+\sqrt{2\pi}/2}{2\sqrt{2\pi}} = 0.75$, so on average $1/0.75 = 1.33$ normal draws per sample. The normalising constant of f^* is never needed, only a bound on f^*/h.

9.3

(a) $f/h = xe^{-(1-\mu)x}/\mu$. Its derivative is zero at $x = 1/(1-\mu)$, giving $k^*(\mu) = e^{-1}/\big(\mu(1-\mu)\big)$.

(b) $\mu(1-\mu)$ is largest at $\mu = \frac{1}{2}$, so $k^* = 4/e = 1.47$ and $P(\text{accept}) = e/4 = 0.68$. (Matches the lecture result $\mu = \lambda/m$.)

(c) If $\mu \ge 1$ the envelope's tail decays at least as fast as the target's (xe^{-x} vs $e^{-\mu x}$): $f/h \to \infty$ as $x \to \infty$, so no finite k exists. The envelope must have heavier tails than the target.

9.4

(a) For $j = 1, \dots, M$: draw $N_j \sim \text{Poisson}(20)$, $p_j \sim \text{Beta}(3,3)$, then $X_j \sim \text{Bin}(N_j, p_j)$. Estimate $P(X \ge 12)$ by the proportion of $X_j \ge 12$.

(b) $\mathrm{E}(X) = \mathrm{E}\big[\mathrm{E}(X \mid N,p)\big] = \mathrm{E}(N)\mathrm{E}(p) = 20 \times 0.5 = 10$. The sample mean of the X_j should be within a few MC standard errors of 10.

(c) $\text{SE} = \sqrt{0.36 \times 0.64/10000} = 0.0048$.

(d) Draw $U \sim U(0,1)$. If $U < 0.3$ return a draw from Exp(1) (e.g. $-\log V$), otherwise return a draw from $\mathrm{N}(5,1)$. This picks the component with the right probability, then samples within it.

Week 10 solutions

10.1

(a) $\pi(\lambda) \propto e^{-\lambda} \times \prod_i \lambda^{y_i} e^{-\lambda} \propto \lambda^9 e^{-4\lambda}$ ($\sum y_i = 9$, $n = 3$).

(b) No: $q(2 \to 4) \ne q(4 \to 2)$, because the proposal's shape and scale depend on the current value. So the proposal ratio must be included.

(c) Posterior ratio: $\pi(4)/\pi(2) = (4/2)^9 e^{-4(4-2)} = 512e^{-8} = 0.1718$.
Proposal densities (the $\Gamma(4)$ cancels): $q(4 \to 2) \propto (4/4)^4 \times 2^3 \times e^{-4\cdot 2/4} = 8e^{-2}$ and $q(2 \to 4) \propto (4/2)^4 \times 4^3 \times e^{-4\cdot 4/2} = 1024e^{-8}$. Ratio $q(4 \to 2)/q(2 \to 4) = e^6/128 = 3.152$.

$$A = \min\left\{1,\ 512e^{-8} \times \frac{e^6}{128}\right\} = \min\{1,\ 4e^{-2}\} = 0.541.$$

10.2

(a) $p(\mu, \tau \mid x) \propto \tau^{1/2} e^{-\tau(1-\mu)^2/2} \times e^{-\mu^2/8} \times \tau^2 e^{-2\tau}$.

(b) q factors as $q(\tau_n \mid \tau_c) \times \mathrm{N}(\mu_n; \mu_c, 1)$. The normal part is symmetric in (μ_c, μ_n), so it cancels. The gamma part does not.

(c) Posterior at $(1,1)$: $1 \times e^{0} \times e^{-1/8} \times 1 \times e^{-2} = e^{-2.125}$. At $(2,2)$: $\sqrt{2} \times e^{-1} \times e^{-4/8} \times 4 \times e^{-4} = 4\sqrt{2}\, e^{-5.5}$. Ratio $= 4\sqrt{2}\, e^{-3.375} = 0.1936$.
Gamma proposal $q(a \to b) \propto (2/a)^2\, b\, e^{-2b/a}$: $q(2 \to 1) = 1 \times 1 \times e^{-1}$, $q(1 \to 2) = 4 \times 2 \times e^{-4}$, ratio $e^3/8 = 2.511$.

$$A = \min\left\{1,\ 4\sqrt{2}\, e^{-3.375} \times \frac{e^3}{8}\right\} = \min\left\{1,\ \tfrac{\sqrt{2}}{2} e^{-0.375}\right\} = 0.486.$$

10.3

(a) $\mathrm{N}(\theta_c, \sigma^2)$ is symmetric: $q(\theta_c \to \theta_n) = q(\theta_n \to \theta_c)$, so $A = \min\{1, \pi(\theta_n)/\pi(\theta_c)\}$.

(b) ESS: $\sigma = 0.1$ gives $5000 \times 0.02/1.98 = 51$; $\sigma = 2.5$ gives $5000 \times 0.28/1.72 = 814$; $\sigma = 25$ gives $5000 \times 0.05/1.95 = 128$. Use $\sigma = 2.5$: the most information per iteration, with an intermediate acceptance rate.

(c) Tiny σ: almost every move is accepted but each is tiny, so the chain crawls (a slowly wandering trace, like a random walk). Huge σ: proposals land far out in the tails and are almost always rejected, so the chain sits still for long stretches (a flat, step-like trace). Either way successive values are nearly equal and highly correlated.

(d) With a symmetric proposal, $A = \min\{1, \pi(\theta_n)/\pi(\theta_c)\}$, so $\pi(\theta) \propto \exp(-|\theta|^3)$.

(e) A depends on π only through the ratio $\pi(\theta_n)/\pi(\theta_c)$, in which the normalising constant $p(x)$ cancels. That constant is exactly the hard integral in Bayes' theorem.

Week 11 solutions

11.1

(a) Keep the factors with μ: $\exp\{-\frac{\tau}{2}\sum(y_i-\mu)^2 - \frac{\kappa_0}{2}(\mu-m_0)^2\} \propto \exp\{-\frac{1}{2}\big[(\kappa_0+n\tau)\mu^2 - 2(\kappa_0 m_0 + \tau\sum y_i)\mu\big]\}$. So

$$\mu \mid \tau, y \sim \mathrm{N}\left(\frac{\kappa_0 m_0 + \tau\sum_i y_i}{\kappa_0 + n\tau},\ \frac{1}{\kappa_0 + n\tau}\right).$$

(b) Keep the factors with τ: $\tau^{n/2} e^{-\tau\sum(y_i-\mu)^2/2} \times \tau^{a-1} e^{-b\tau} = \tau^{a+n/2-1} e^{-\tau[b+\sum(y_i-\mu)^2/2]}$, so

$$\tau \mid \mu, y \sim \mathrm{Gamma}\left(a + \frac{n}{2},\ b + \frac{1}{2}\sum_i (y_i-\mu)^2\right) \quad \text{(shape, rate)}.$$

(c) Set $(\mu^{(0)}, \tau^{(0)})$. For $t = 1, \ldots, T$: draw $\mu^{(t)}$ from $p(\mu \mid \tau^{(t-1)}, y)$, then $\tau^{(t)}$ from $p(\tau \mid \mu^{(t)}, y)$, always conditioning on the latest values. Discard the first B draws as burn-in; estimate $P(\mu > 2 \mid y)$ by the proportion of the remaining $\mu^{(t)}$ above 2.

(d) $\tau = 2$: precision $1 + 4 \times 2 = 9$, mean $2 \times 10/9 = 2.22$, so $\mu \mid \tau = 2, y \sim \mathrm{N}(2.22,\ 1/9)$.
$\mu = 2.5$: $\sum(y_i - 2.5)^2 = \sum y_i^2 - 2(2.5)\sum y_i + 4(2.5)^2 = 30 - 50 + 25 = 5$, so $\tau \mid \mu = 2.5, y \sim \mathrm{Gamma}(3,\ 3.5)$.

(e) With independent priors, the joint posterior is not a standard family: the marginal of τ involves $1/\sqrt{\kappa_0 + n\tau}$ and a non-gamma term, so it cannot be sampled directly. The normal–gamma prior is conjugate: $p(\tau \mid y)$ is Gamma and $p(\mu \mid \tau, y)$ is Normal, so you can sample τ and then μ exactly with no Markov chain.

11.2

(a) The priors on λ_1, λ_2 and the uniform prior on m are constant in m, so

$$\text{joint} \propto \prod_{i \le m} \lambda_1^{y_i} e^{-\lambda_1} \prod_{i > m} \lambda_2^{y_i} e^{-\lambda_2} = \lambda_1^{S_m} e^{-m\lambda_1}\, \lambda_2^{T - S_m} e^{-(5-m)\lambda_2},$$

with $T = \sum y_i$. Drop $\lambda_2^T e^{-5\lambda_2}$ (no m): $\propto (\lambda_1/\lambda_2)^{S_m} e^{-m(\lambda_1 - \lambda_2)}$.

(b) $\lambda_1/\lambda_2 = 3.75$, $\lambda_1 - \lambda_2 = 2.2$, $S = (3, 7, 8, 8)$. Weights $3.75^3 e^{-2.2}$, $3.75^7 e^{-4.4}$, $3.75^8 e^{-6.6}$, $3.75^8 e^{-8.8}$ are proportional to $(0.0144,\ 0.315,\ 0.131,\ 0.0145)$; normalising gives

$$P(m = 1, 2, 3, 4 \mid \cdot) = (0.030,\ 0.663,\ 0.276,\ 0.031).$$

(c) m has finite support, so compute the four normalised probabilities and draw from that discrete distribution (one uniform and the cdf ladder of Week 8, or `sample(1:4, 1, prob = w)`).

11.3

Each update moves only parallel to an axis, and the conditional SD is $\sqrt{1 - \rho^2} = 0.31$, much narrower than the marginal SD of 1. Along the long diagonal of the ellipse the chain takes tiny zig-zag steps, so successive draws are highly autocorrelated and it explores slowly. Remedies: update (X_1, X_2) jointly (blocking), reparametrise to less correlated coordinates (e.g. $X_1 + X_2$ and $X_1 - X_2$), or use an MH proposal aligned with the correlation.

11.4

(a) α: $\hat{R} = 1.31/1.02 = 1.28 > 1.05$, not converged (the chains disagree). β: $1.03/1.01 = 1.02 < 1.05$, acceptable.

(b) If all chains start in the same region, they can agree with each other while all missing another part of the posterior. Spread-out starts that come together are real evidence of convergence.

(c) ESS $\approx 15000 \times (1 - 0.6)/(1 + 0.6) = 3750$, just below 4000: run longer.

(d) Burn-in: early iterations still depend on the starting value and are not yet draws from the posterior, so they would bias estimates. Thinning (keeping every k-th draw) reduces autocorrelation in what you store and saves memory, but throws information away; the ESS of the full chain is never smaller than that of the thinned one.

(e) The three α traces sit at different levels or wander slowly without overlapping well ("not mixing"), whereas well-converged traces look like overlapping fuzzy horizontal bands.

Week 12 solutions

12.1

Start from

$$\log p(\mu, \tau, y) = \frac{n}{2} \log \tau - \frac{\tau}{2} \sum_i (y_i - \mu)^2 - \frac{\kappa_0 \mu^2}{2} + (a - 1) \log \tau - b\tau + \text{const.}$$

(a) Terms with μ, τ replaced by $\mathrm{E}[\tau]$: $-\frac{\mathrm{E}[\tau]}{2} \sum (y_i - \mu)^2 - \frac{\kappa_0 \mu^2}{2} = -\frac{1}{2}\big[(\kappa_0 + n\mathrm{E}[\tau])\mu^2 - 2\mathrm{E}[\tau] \sum y_i\, \mu\big] + \text{const.}$ So

$$q_\mu^* = \mathrm{N}\left(\frac{\mathrm{E}[\tau] \sum y_i}{\kappa_0 + n\mathrm{E}[\tau]},\ \frac{1}{\kappa_0 + n\mathrm{E}[\tau]} \right).$$

(b) Terms with τ, expectation over μ: $(a+n/2-1)\log\tau - \tau\big[b+\frac{1}{2}\sum \mathrm{E}_\mu(y_i-\mu)^2\big]$. Since $\mathrm{E}_\mu(y_i-\mu)^2 = (y_i-\mathrm{E}[\mu])^2 + \mathrm{V}[\mu]$:

$$q^*_\tau = \mathrm{Gamma}\left(a+\frac{n}{2},\ b+\frac{1}{2}\Big[\sum_i (y_i-\mathrm{E}[\mu])^2 + n\,\mathrm{V}[\mu]\Big]\right).$$

(c) With $\mathrm{E}[\tau]=1$: $q_\mu = \mathrm{N}(10/5,\ 1/5) = \mathrm{N}(2,\ 0.2)$. Then $\sum(y_i-2)^2 = 30-40+16 = 6$, plus $n\mathrm{V} = 0.8$, total 6.8: $q_\tau = \mathrm{Gamma}(3,\ 1+3.4) = \mathrm{Gamma}(3,\ 4.4)$, so $\mathrm{E}[\tau] = 0.682$. (Iterating converges to $q_\mu = \mathrm{N}(1.74,\ 0.305)$, $q_\tau = \mathrm{Gamma}(3,\ 5.28)$.)

(d) Same family and the same formulas, but the Gibbs conditional uses the current *random draw* of τ, whereas CAVI uses the fixed number $\mathrm{E}[\tau]$ (and $\mathrm{V}[\mu]$ enters q_τ). Gibbs produces random samples; CAVI is a deterministic optimisation that returns distributions.

12.2

(a) $\log p = y\log\lambda - \lambda + \big[2\log\beta + \log\lambda - \beta\lambda\big] - \beta + \text{const} = (y+1)\log\lambda - \lambda(1+\beta) + 2\log\beta - \beta + \text{const}$.

(b) In λ: $(y+1)\log\lambda - \lambda(1+\mathrm{E}[\beta])$, the $\mathrm{Gamma}\big(y+2,\ 1+\mathrm{E}[\beta]\big)$ kernel. In β: $2\log\beta - \beta(1+\mathrm{E}[\lambda])$, the $\mathrm{Gamma}\big(3,\ 1+\mathrm{E}[\lambda]\big)$ kernel.

(c) $\mathrm{E}[\lambda] = 6/(1+\mathrm{E}[\beta])$ and $\mathrm{E}[\beta] = 3/(1+\mathrm{E}[\lambda])$. Cross-multiplying and subtracting: $\mathrm{E}[\lambda]-\mathrm{E}[\beta] = 3$. Substituting: $\mathrm{E}[\beta]^2 + 4\mathrm{E}[\beta] - 3 = 0$, so $\mathrm{E}[\beta] = \sqrt{7}-2 = 0.646$ and $\mathrm{E}[\lambda] = \sqrt{7}+1 = 3.646$. Hence

$$q^*_\lambda = \mathrm{Gamma}\big(6,\ \sqrt{7}-1 = 1.646\big), \qquad q^*_\beta = \mathrm{Gamma}\big(3,\ \sqrt{7}+2 = 4.646\big).$$

(d) No. Integrating out β exactly gives $p(\lambda \mid y) \propto \lambda^5 e^{-\lambda}/(1+\lambda)^3$, which is not a gamma. The mean-field q ignores the dependence between λ and β, so it is only the closest product-form approximation in KL.

12.3

(a) $\mathrm{KL}\big(q \,\|\, p(\theta \mid y)\big) = \mathrm{E}_q\big[\log q - \log p(\theta, y)\big] + \log p(y) = -\mathrm{ELBO}(q) + \log p(y)$. $\log p(y)$ does not depend on q, so maximising the ELBO minimises the KL. The ELBO uses only the joint $p(\theta, y)$, which we can write down.

(b) $\mathrm{KL} \geq 0$, so $\log p(y) \geq \mathrm{ELBO}(q)$: it is a lower bound on the log evidence (marginal likelihood).

(c) $\mathrm{KL}(q \,\|\, p) = \mathrm{E}_q[\log q/p]$ is huge wherever q puts mass where p is small, so the optimal q avoids those regions and sits inside the posterior. With a factorised q it cannot capture correlation, and it ends up too narrow. VI is preferable for very large datasets or complex models where MCMC is too slow, or when a quick approximate answer is enough.

(d) No. It is a restriction on the approximating family only; the model and the exact posterior can have any dependence between μ and τ. It limits how well q can match the posterior.